

Oct 30

Midterm -2 on Nov 6, Wednesday

↳ T.D.P.T, FGR, dipole transitions

- 1 page notes.

Today : Review Scattering

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Warm-up:

Given:

T.D.S.E $i\hbar \frac{d}{dt} |\psi(t)\rangle = (H_0 + V(t)) |\psi(t)\rangle$

Interaction picture:

$$|\psi_I(t)\rangle = e^{iH_0 t/\hbar} |\psi(t)\rangle$$

Find : $i\hbar \frac{d}{dt} |\psi_I(t)\rangle = ? |\psi_I(t)\rangle$

$$\Rightarrow \text{LHS} = i\hbar \frac{d}{dt} |\psi_I(t)\rangle$$

$$= i\hbar \frac{d}{dt} \left(e^{iH_0 t/\hbar} |\psi(t)\rangle \right)$$

* clear
up the
interaction
picture
& Schrödinger
picture
quantities

$$= i\hbar \left[e^{iH_0 t/\hbar} \frac{d}{dt} (|\psi(t)\rangle) + |\psi(t)\rangle \frac{iH_0 e^{iH_0 t/\hbar}}{\hbar} \right]$$

$$= e^{iH_0 t/\hbar} \underbrace{i\hbar \frac{d}{dt} |\psi(t)\rangle}_{\rightarrow H_0 + V(t) |\psi(t)\rangle} = H_0 e^{iH_0 t/\hbar} |\psi(t)\rangle$$

(given T.D.S.E.)

$$= e^{iH_0 t/\hbar} (H_0 + V(t)) |\psi(t)\rangle - H_0 e^{iH_0 t/\hbar} |\psi(t)\rangle$$

$$= e^{iH_0 t/\hbar} \cancel{H_0} |\psi(t)\rangle + e^{iH_0 t/\hbar} V(t) |\psi(t)\rangle - \cancel{H_0 e^{iH_0 t/\hbar} |\psi(t)\rangle}$$

$$= \underbrace{e^{iH_0 t/\hbar} V(t) e^{-iH_0 t/\hbar}}_{V_I(t)} \cdot \underbrace{e^{iH_0 t/\hbar} |\psi(t)\rangle}_{|\psi_I(t)\rangle}$$

$$i\hbar \frac{d}{dt} |\psi_I(t)\rangle = V_I(t) |\psi_I(t)\rangle$$

Solve :

$$\frac{d}{dt} |\psi_I(t)\rangle = \frac{1}{i\hbar} V_I(t) |\psi_I(t)\rangle$$

integrate.

$$\begin{aligned} \int_{t_0}^t dt' \frac{d}{dt'} |\psi_I(t')\rangle &= |\psi_I(t)\rangle - |\psi_I(t_0)\rangle \quad \text{fundamental thm of calculus.} \\ &= \frac{1}{i\hbar} \int_{t_0}^t V_I(t') |\psi_I(t')\rangle dt' \quad \text{LHS integral} \\ &\quad \text{integral of RHS.} \end{aligned}$$

$$|\psi_I(t)\rangle = |\psi_I(t_0)\rangle + \frac{1}{i\hbar} \int_{t_0}^t dt' V_I(t') |\psi_I(t')\rangle$$

$$\left(= |\psi_I(t_0)\rangle + \frac{1}{i\hbar} \int \dots \right) \quad o(v)$$

$$|\psi_I(t)\rangle = |\psi_I(t_0)\rangle + \frac{1}{i\hbar} \int_{t_0}^t dt' V_I(t') |\psi_I(t_0)\rangle$$

$$+ \left(\frac{1}{i\hbar} \right)^2 \int_{t_1}^t dt_2 \int_{t_0}^{t_2} dt_1 V_I(t_2) V_I(t_1) |\psi_I(t_0)\rangle$$

+ ...

$$\equiv U_I(t, t_0) |\psi_I(t_0)\rangle$$

$$U_I(t, t_0) = \mathbb{I} + \frac{1}{i\hbar} \int_{t_0}^t dt' V_I(t') + \dots$$

- Initial state : $|\psi_I(t_0)\rangle = |i\rangle$

$$H_0 |i\rangle = \hbar \omega_i |i\rangle$$

$$|\psi_I(t)\rangle = |i\rangle + \frac{1}{i\hbar} \int_{t_0}^t dt' V_I(t') + \dots$$

$$\langle n | \psi_I(t) \rangle = \delta_{ni} + \frac{1}{i\hbar} \int_{t_0}^t dt' \langle n | V_I(t') | i \rangle$$

$$= \delta_{ni} + \frac{1}{i\hbar} \int_{t_0}^t dt' \langle n | e^{iH_0 t'/\hbar} V(t') e^{-iH_0 t'/\hbar} | i \rangle$$

$$= \delta_{ni} + \frac{1}{i\hbar} \int_{t_0}^t dt' e^{i(\omega_n - \omega_i)t'} \langle n | V(t') | i \rangle + \dots$$

$$c_n^{(1)} = \frac{1}{i\hbar} \int_{t_0}^t dt' e^{i\omega_n t'} V_{ni}(t')$$

$$|\psi_I(t)\rangle = \sum_n |n\rangle \underbrace{\langle n | \psi_I(t) \rangle}_{c_n(t)} = \sum_n c_n(t) |n\rangle$$

$$c_n(t) = c_n^{(0)} + c_n^{(1)}(t) + \dots$$

$$|\psi(t)\rangle = e^{-iH_0 t/\hbar} |\psi_I(t)\rangle = \sum_n c_n(t) e^{-i\omega_n t} |n\rangle$$

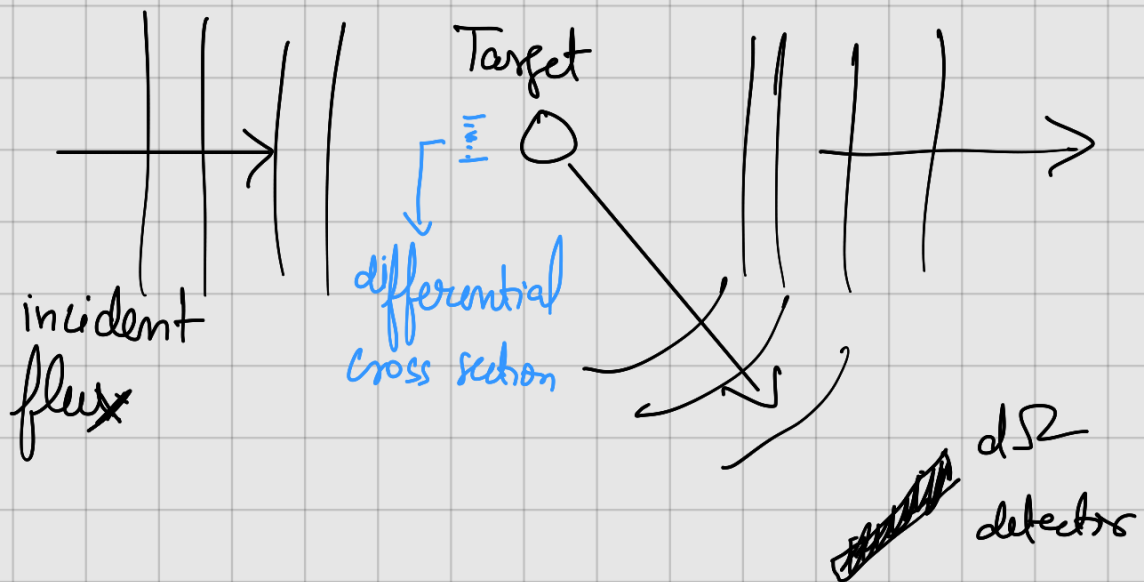
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Scattering Theory

- Particle in 3D (wave), no spin
- target (infinitely massive) at $\vec{r} = 0$

same theory works for two particles with

record
in center
of mass
frame.



incident flux $F = \frac{\# \text{ particles}}{\text{Area} \times \text{time}}$

rate of detection $\leftarrow$
detect $\frac{dR}{d\Omega} = \frac{\# \text{ particles}}{\text{time} \times d\Omega}$

- cross-section :

$$\frac{dR}{d\Omega} = F \left(\frac{d\sigma}{d\Omega} \right)$$

$\frac{d\sigma}{d\Omega}$ differential cross-section.
(essentially the "area"
of the target)

$$\Rightarrow \frac{d\sigma}{d\Omega} = \frac{1}{F} \frac{dR}{d\Omega}$$

rate can be calculated using Fermi's Golden rule.

FGR gives the transition rates.

$$H = \frac{\vec{p}^2}{2m} + V(\vec{r}) = H_0 + V$$

$$H_0 = \frac{\vec{p}^2}{2m} \quad (\text{free particle})$$

- use periodic boundary conditions in box of dimension $L \times L \times L$

$$\phi_{\vec{k}}(\vec{r}) = \frac{1}{\sqrt{L^3}} e^{i\vec{k} \cdot \vec{r}}$$

$$\vec{k} \approx \frac{2\pi}{L} (n_x, n_y, n_z)$$

- initial state: $\phi_i(\vec{r}) = \frac{1}{\sqrt{L^3}} e^{i\vec{k}_i \cdot \vec{r}}$

wave vector pointing toward the target

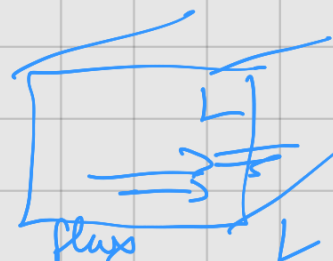
- flux: $F = \frac{1 \text{ particle}}{L^2 \cdot \frac{L}{v}}$

area

←

→ time

(dist. / speed)



$$= \frac{V}{L^3} = \frac{\hbar k_i / m}{L^3}$$

* final state would have a wave vector pointing towards the detector, so that we can use F.G.R to calculate the transition b/w these initial and final states.

F.G.R rate into $d\Omega$

$$dR = \frac{2\pi}{\hbar} |\langle k_f | V | k_i \rangle|^2 g_f(E_i)$$

$$g_f = \frac{m L^3 k}{8 \pi^3 \hbar^2} d\Omega \quad (|\vec{k}_i| = |\vec{k}_f| = k)$$

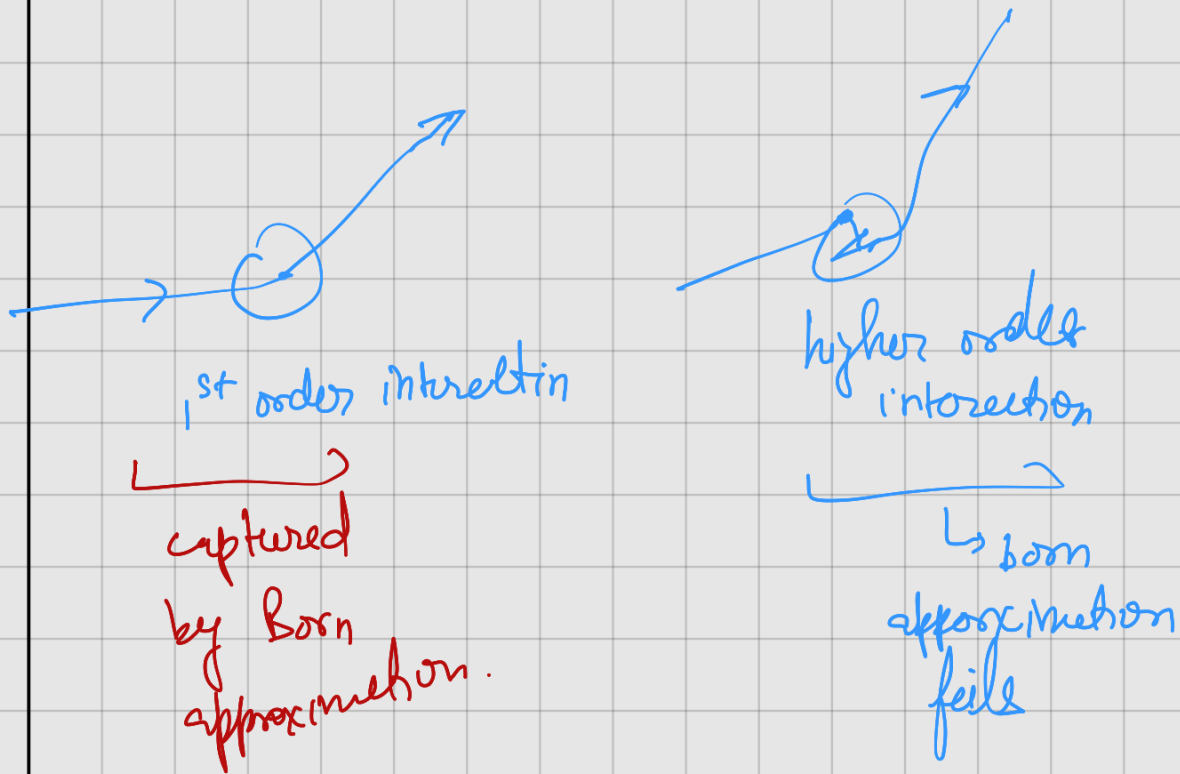
$$\frac{dR}{d\Omega} = \frac{2\pi}{\hbar} |V_{k_i}|^2 \frac{m L^3 k}{8 \pi^3 \hbar^2} = \frac{m L^3 k |V_{f_i}|^2}{(2\pi)^2 \hbar^3}$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{F} \frac{dR}{d\Omega} = \frac{L^3 m}{\hbar k} \frac{m L^3 k}{(2\pi)^2 \hbar^3} \left| \int d^3 r \frac{1}{L^3} e^{-i\vec{k}_f \cdot \vec{r}} V(\vec{r}) e^{i\vec{k}_i \cdot \vec{r}} \right|^2$$

Born Approximation

$$\frac{d\sigma}{d\Omega} = \left(\frac{m}{2\pi \hbar^2} \right)^2 \left| \int d^3 r e^{i(\vec{k}_i - \vec{k}_f) \cdot \vec{r}} V(\vec{r}) \right|^2$$

* gives differential cross section for every input potential $V(\vec{r})$



* $\Rightarrow$ for Born approximation to be valid, interaction order (perturbation potential) can't be high.

